SECTION I (1 mark per question):

1. What is the output of the following program?

```
#include <stdio.h>

int
wilma (int x)
{
        x = x + 3;
        if (x < 5)
                x = 7;
        return (x);
}
int
main (void)
{
        int x=1;
        x = wilma (x);
        printf ("%d", x);
        return (0);
}
```

a) 1 b) 3 c) 4 d) 7
e) none of these

2. What is the output of the following program?

```
#include <stdio.h>
int merlin (int y)
{
        y = y * 2;
        return (y);
}

int arthur (int x)
{
        x = merlin(x) * 2;
        return (x);
}
int
main (void)
{
        int n = 5;
        n = arthur (n) + 20;
        printf ("%d", n);
        return (0);
}
```

a) 10 b) 20 c) 40 d) 60
e) none of these

3. In C, if an array is declared as **double array [26];**, how many numbers can you put in it?

a) 50 b) 25 c) 24
d) as many as you need e) none of these

4. If a string is declared as **char capital[20];**, and the statement **strcpy (capital, "Edmonton");** is executed, what is the value of **capital[8]**?

a) 'E' b) '0' c) '\n' d) 'n'
e) none of these

5. If a function is defined as
double andromeda (double z, int a) {return (z+a);}, what is the value of **andromeda (3, 2.5)**?

a) 2.5 b) 3 c) 5 d) 5.0 e) 5.5

6. What is the equivalent of
char city[ ] = {'B', 'o', 's', 't', 'o', 'n', '\0'};?

a) *char city[ ] = 'Boston';*
b) *char city[ ] = "Boston";*
c) *char city[ ] = "Boston\0";*
d) *char city[ ] = Boston;*
e) *none of the above*

7. Which of the following functions would be recommended to read an entire sentence as input?

a) fgets b) gets c) scanf d) input
e) none of these

8. Identify the correct structure definition.

a) def struct gandalf {int frodo; double bilbo; char saruman[20];}
b) structure gandalf {int frodo; double bilbo; char saruman[20];}
c) struct gandalf {int frodo; double bilbo; char saruman[20];}
d) typedef gandalf {int frodo; double bilbo; char saruman[20];}
e) none of the above

9. Assuming a declaration of char dumbledore[10];, which of the following statements is correct?

a) scanf ("%s", &dumbledore);
b) fscanf ("%c", dumbledore);
c) scanf ("%s", dumbledore);
d) fgets (&dumbledore);
e) none of the above

10. What is the proper loop definition to travel through an entire linked list? (**head** is an existing pointer to the first node of the list)

a) for (i=0; head<size; ++i)
b) for (i=0; head!=NULL; ++i)
c) for (i=head; i!=NULL; i=i->next)
d) for (i=head; i!=NULL; ++i)
e) none of the above

11. Identify the correct structure definition that allows the structure to be used as nodes of a linked list.

a) list gandalf {int frodo; double bilbo; gandalf bree;}
b) struct gandalf {int frodo; double bilbo; struct gandalf *bree;}
c) struct gandalf {int frodo; double bilbo; struct gandalf bree;}
d) struct gandalf {int frodo; double bilbo; gandalf* bree;}
e) none of the above

12. We use atoi to convert a string into a number. What is the value of atoi("3.4e2");?

a) 3.4 b) 3.40 c) 3 d) 340.0
e) none of the above

13. What is the output of the following program?

```
#include <stdio.h>

typedef struct fred
{
    int barney[2];
    struct fred* next;
}fred;


int
main (void)
{
        fred f1, f2, f3, f4;
        fred* x;

        f4.next = &f2;
        f1.barney[0]=3; f1.barney[1]=5; f2.barney[0]=7; f2.barney[1]=6;
        f2.next = &f1;
        f3.barney[0]=2; f3.barney[1]=4; f4.barney[0]=9; f4.barney[1]=8;
        f1.next = NULL;
        x = &f4;

        while (x != NULL)
        {
                printf ("%d  ", x -> barney[0]);
                x = x -> next;
        }
        return (0);
}
```

a) 5 6 4 8
b) 8 6 5
c) 9 7 3
d) 3 7 2 9
e) None of the above

SECTION II (1 mark per question):
Identify the correct missing line in the following C programs. The missing line must make the program produce results (no compile or runtime error) consistent with the labelling in the output statements. The position of the missing line is indicated by a ▶ in the left margin.

14.

```
#include <stdio.h>
int
main (void)
{
▶
      x[3] = 32.5;
      printf ("The temperature is %lf", x[3]);
      return (0);
}
```

a) double x[];
b) double x[4];
c) double* x[10];
d) double x[3];
e) none of the above

15.

```
#include <stdio.h>
int dino (int y[])
{
      int x;
      x = y[0];
      return (x);
}
int
main (void)
{
      int array[3] = {1,2,3}, n;
      /* prints out the first element (cell) of the array */
▶
      printf("%d\n", n);
      return (0);
}
```

a) n = dino (array);
b) n = y[0];
b) n = array[1];
d) dino (&n, array);
e) dino (n, array[0]);

Last exam question:

11. The ratio between successive speeds of a six-speed gearbox (assuming that the gears are evenly spaced to allow for whole teeth) is

$$\sqrt[5]{M/m}$$

where M is the maximum speed in revolutions per minute and m is the minimum speed. Write a function `speeds_ratio` that calculates this ratio for any maximum and minimum speeds. Write a main function that prompts for maximum and minimum speeds (rpm), calls `speeds_ratio` to calculate the ratio, and displays the results in a sentence of the form

```
The ratio between successive speeds of a six-speed gearbox
with maximum speed __________ rpm and minimum speed __________
rpm is __________.
```

Question 17 (6 marks):

Please read the constraints carefully. No marks will be awarded if the requirements are not followed.

Below, you are to complete a main function, preceded by a **void** function (**no** return statement) named **moose**. **moose** adds three INTEGER values to produce their sum. The function **moose** may have only THREE parameters, the FIRST and THIRD of which, named **buffalo** and **other**, must be of type **int**. The function has only a SINGLE line in its body.

The main function must call the function **moose** so that the function **moose** (in the program below) causes the printed output from the main program to be 99. There may not be any calculation at all in the main function. No given statements may be modified. *No additional statements may be added.*

There may be no printed output from ***moose****, and line c may not involve any printing.* You may write only the three lines: a) complete the function header, b) give a single statement in the function body, c) a statement in the main function to invoke (call) the function moose. (The letterings are indicated as comments)

```
#include <stdio.h>

void
moose ( ______________________________) /* a */
{
    ___________________________________ /* b */
}

int
main (void)
{
    int tiny;
    int abalone, tiny_abalone;
    abalone = 11;
    tiny = 22;
    tiny_abalone = 66;
    ___________________________________ /* c */
    printf( "%d \n" , tiny_abalone);
}
```